CrimeNexus

Final Feature & Implementation Plan

AI-Powered Criminal Network Analysis System — SIH26189
Where Every Clue Connects

Purpose: A secure, region-based investigator web platform that converts fragmented investigation data into an explainable criminal knowledge graph, supports cross-case discovery and temporal reconstruction, provides evidence-backed AI assistance, and maintains tamper-evident evidence integrity.

1. Region-Based Secure Login & Access Control

What it does

- Police stations belonging to one region/jurisdiction use a common regional login.
- After authentication, investigators can access only cases, evidence, reports and investigation data belonging to that region.
- One region cannot directly access another region's investigation files.
- Uploads, evidence access, verification and investigation actions are associated with the region and authenticated session.

Technology

Frontend: React / Next.js **Backend:** FastAPI + Python **Database:** PostgreSQL **Storage:** MinIO
Authentication: JWT **Password Security:** Argon2 **Authorization:** Region/Jurisdiction-based access control

Implementation

```
Regional Login
↓
JWT Authentication
↓
Identify Region
↓
Region-based Authorization
↓
Cases / Evidence / Data filtered by region_id
```

2. Investigation Data Upload / Add Data

What it does

- Upload FIRs and case reports.
- Upload CDR records.
- Upload financial transaction records.
- Upload vehicle, device and cyber records.
- Upload other investigation evidence/documents.

Technology

React / Next.js, Tailwind CSS, FastAPI, Python, Pandas, PyMuPDF, PaddleOCR, PostgreSQL, MinIO, REST APIs.

Implementation

```
Upload File
↓
Validate File
```

```
↓
Calculate SHA-256
↓
Store Original File → MinIO
↓
Extract Data
■■■ PDF → PyMuPDF
■■■ Scanned PDF → PaddleOCR
■■■ CSV → Pandas
↓
Extract / Normalize / Resolve Entities
↓
Store Structured Data
↓
Update Knowledge Graph
```

3. Cases

What it does

- View all cases belonging to the investigator's region.
- Search by Case ID, person, phone, account, vehicle or IP.
- Open a complete case workspace.
- View evidence, entities, graph, timeline, similar cases and investigation status.
- Start Investigation Reconstruction or ask the Copilot questions.

Technology

React / Next.js, Tailwind CSS, FastAPI, PostgreSQL, Neo4j, MinIO.

Example Case View

```
CASE #018
Financial Fraud Investigation
Status: Active
Region: Region A

Evidence: 8 | People: 15 | Phones: 8
Accounts: 6 | Vehicles: 4

[Open Case] [Investigate]
```

4. Criminal Knowledge Graph

What it does

Represents people, cases, phones, bank accounts, transactions, vehicles, devices, IPs, emails, domains, locations and evidence as interconnected nodes rather than isolated records.

Technology

Neo4j — graph database **Cypher** — graph queries **Neo4j GDS / NetworkX** — analytics **Cytoscape.js** — visualization **FastAPI + Python** — backend

Implementation

```
PostgreSQL Data + Extracted Entities + Resolved Entities
↓
Neo4j
↓
Knowledge Graph
↓
Cytoscape.js
↓
Interactive Case Network
```

5. Expandable Entity Cards

What it does

Clicking any graph node opens a detailed investigator card rather than only showing the entity name.

- Person/entity ID and associated cases.
- Phones, accounts, vehicles and devices.
- Known locations and related entities.
- Network role and timeline.
- Evidence supporting relationships and confidence.
- Potentially relevant legal provisions and flagged contradictions.

Actions: View Evidence • View Source • View Timeline • Expand Network • Open Related Case

Technology

React / Next.js, FastAPI, Neo4j, PostgreSQL, MinIO, Cytoscape.js.

6. AI Investigation Reconstruction — Signature Feature

What it does

The investigator clicks **START INVESTIGATION**. CrimeNexus progressively reconstructs the investigation chronologically. As evidence-backed events are introduced, narration is displayed and the live graph grows in real time.

Core flow

```
ADD DATA
↓
START INVESTIGATION
↓
AI NARRATION
↓
LIVE GRAPH BUILDS
↓
ENTITY CARDS APPEAR
↓
CROSS-CASE CONNECTIONS
↓
EVIDENCE / CONFLICTS / GAPS
↓
INVESTIGATION SUMMARY
```

Example

Event 1: “On 9 June, Person A's account received ■1 crore.” The Person and Account nodes appear.

Event 2: “On 11 June, Person B transferred ■10 crore through five accounts.” Additional nodes and transaction relationships appear.

Event 3: A later relationship connects the activity to another case. The graph expands and the narrator explains the evidence-backed connection.

Technology

React / Next.js, Cytoscape.js, FastAPI, Python, Neo4j, PostgreSQL, LLM/Llama or suitable LLM API, dateparser. Optional: Text-to-Speech.

AI safety / evidence rule

The LLM should receive retrieved investigation facts and narrate them. It should not invent events, motives, guilt or unsupported legal conclusions. The database and source evidence remain the source of truth.

7. Investigation Gaps & Evidence Contradiction Detection

What it does

Flags incomplete evidence or conflicting records during investigation reconstruction.

- Example: a transaction proves money moved, but the recipient account owner is not yet established.
- Example: two records place a vehicle/device in different locations at the same time.
- CrimeNexus flags the conflict and tells the investigator what should be reviewed; it does not decide which source is correct.
- Suggested review actions can include checking additional CDRs, vehicle records, device evidence or source documents.

Technology

Python, PostgreSQL, Neo4j, spaCy/Transformers, dateparser, FastAPI. Initial implementation can be rule-based with NLP extraction; custom model training is not required.

8. Similar Cases / M.O. Search

What it does

Finds historically similar cases based on the semantic pattern of an incident, even when the cases do not share an obvious person, phone or account.

Technology

Sentence Transformers, FAISS/Qdrant, Python, PostgreSQL, FastAPI, LLM for explanation.

Example

```
Case #018
Financial transfer → multiple intermediary accounts → incident
↓
Similarity Search
↓
Case #041
Similar transaction pattern → multiple intermediary accounts
```

9. AI Investigation Copilot

What it does

A natural-language assistant that investigators can use to ask questions about any case, entity, relationship, timeline or cross-case connection.

- “Is Case 18 related to Case 41?”
- “Why was Person B highlighted?”
- “Show everyone connected to Account X.”
- “What happened before the incident?”
- “Why might this relationship exist?”

Implementation

```
Investigator Question
↓
LLM
↓
Structured Graph / DB Query
↓
Query Validation
↓
Neo4j / PostgreSQL
↓
Evidence Retrieval
```

```
↓
LLM Explanation
↓
Evidence-backed Answer + Visual Result
```

Technology: React / Next.js, FastAPI, LLM/Llama, Neo4j + Cypher, PostgreSQL, Sentence Transformers, FAISS/Qdrant.

Example output: Case 18 ↔ Case 41 graph comparison + supporting records + [View Case 18] [View Case 41] [View Graph Path] [View Evidence].

10. Network Intelligence

What it does

- Degree centrality — direct connectivity.
- Betweenness centrality — bridge/broker-like structural position.
- PageRank — structural importance.
- Community detection — densely connected groups.
- Connected components — separated network portions.
- Optional anomaly/statistical analysis.

Technology

Neo4j Graph Data Science, NetworkX, Python, Scikit-learn, FastAPI.

Interpretation: Use evidence-neutral language such as “high network-brokerage characteristics; may warrant further review,” not “mastermind” or “guilty.”

11. Timeline / Temporal Investigation

What it does

Shows when relationships and events happened and how the network evolved. A date-range slider can filter the graph to the state of the network during a selected period.

Technology

React / Next.js, Cytoscape.js, Neo4j, PostgreSQL, Python, dateparser, FastAPI.

```
09 JUN → Account transaction
11 JUN → Communication / transfer
13 JUN → Additional transaction
15 JUN → Incident
↓
Timeline + Live Graph
```

12. Explainable Evidence Panel

What it does

Answers: **“Why did CrimeNexus find this?”** Every important entity, relationship and analytical finding is traceable to its supporting source record.

```
Person B → Account Y
Source: TXN_552
Record: #552
Timestamp: 07 Aug 2026
Confidence: 91%
Extraction: Transaction Dataset

[Open Source Record]
```

Technology

React, FastAPI, Neo4j, PostgreSQL, MinIO, PyMuPDF, PaddleOCR.

13. SHA-256 + Blockchain Evidence Integrity

What it does

Provides the main Cybersecurity + Blockchain layer. Actual evidence files remain in secure off-chain storage; cryptographic hashes and relevant audit information are recorded on a permissioned ledger.

```
Evidence File
↓
SHA-256 Hash
↓
Hash + Metadata + Audit Event
↓
Hyperledger Fabric
```

Verification

```
Current Evidence
↓
Generate SHA-256 Again
↓
Compare With Recorded Hash
↓
MATCH → Evidence Integrity Verified
MISMATCH → Integrity Warning
```

Technology

Python hashlib, SHA-256, MinIO, PostgreSQL, Hyperledger Fabric, FastAPI, React, JWT, Argon2.

14. Audit Trail

What it does

- Evidence uploaded.
- Evidence accessed.
- Evidence verified.
- Investigation record created.
- Investigation action performed.

This provides a tamper-evident history of important evidence and investigation actions.

15. Legal Intelligence + Digital Evidence Dossier

What it does

After investigation reconstruction, CrimeNexus can identify **potentially relevant legal provisions** based on extracted facts and evidence, and assist in preparing an electronic evidence dossier/certificate.

Evidence dossier can include

- Case ID and Evidence ID.
- Source/device information.
- Evidence description.
- SHA-256 hash and algorithm.
- Timestamp and chain-of-custody metadata.
- Integrity verification status.
- Authorized investigator information.

Important: CrimeNexus assists with preparation and review. It should not claim to independently determine guilt, legal liability or court admissibility.

16. Overall Website Architecture

```
REGION LOGIN
↓
REGION DASHBOARD
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ADD DATA ■ CASES ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Extraction Open Case
↓ ↓
■■■■■■■■■→ Knowledge Graph
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓ ↓
Graph Timeline Similar Cases
↓
INVESTIGATION MODE
↓
AI NARRATION
↓
LIVE GRAPH
↓
EXPANDABLE CARDS
↓
EVIDENCE / GAPS / CONFLICTS
↓
COPILOT
↓
INVESTIGATION SUMMARY
↓
LEGAL INTELLIGENCE
↓
DIGITAL EVIDENCE DOSSIER
↓
SHA-256 + BLOCKCHAIN
↓
AUDIT TRAIL
```

17. Final Technology Stack

Area	Technology
Frontend	React / Next.js / Tailwind CSS
Graph Visualization	Cytoscape.js
Backend	FastAPI + Python
Graph Database	Neo4j + Cypher
Graph Analytics	Neo4j GDS + NetworkX
Structured Database	PostgreSQL
Evidence Storage	MinIO
PDF Extraction	PyMuPDF
OCR	PaddleOCR
NLP	spaCy / Hugging Face
Semantic Retrieval	Sentence Transformers + FAISS/Qdrant
Statistics / Anomaly	Scikit-learn
Date Processing	dateparser

Area	Technology
AI Copilot / Narration	Llama or suitable LLM API
Integrity	SHA-256 / Python hashlib
Blockchain	Hyperledger Fabric
Security	JWT + Argon2
Deployment	Docker
Version Control	Git + GitHub

18. Synthetic Dataset Strategy

Use one interconnected synthetic CrimeNexus investigation dataset rather than unrelated datasets. This dataset should support the entire demo.

- Cases — Case ID, title, date, location, category.
- FIRs/reports — fictional narratives containing entities and events.
- People — fictional names/IDs and case associations.
- CDRs — caller, receiver, timestamp, duration.
- Transactions — sender, receiver, amount, timestamp, transaction ID.
- Vehicles — vehicle ID, registration, person, location, timestamp.
- Devices — device ID, person, timestamps.
- Cyber indicators — IP, email, domain, device, timestamp.
- Locations — address/coordinates and case/entity references.
- Evidence — document ID, source, timestamp, hash metadata.

Dataset design should include several apparently disconnected cases, hidden connections distributed across records, at least one bridge/broker-like person, two or three communities, timestamped event sequences, source records for important relationships, and a ground-truth file describing intentionally created hidden relationships.

19. Team Development Plan

Person 1 — Repository + Synthetic Dataset

- Create the main GitHub repository.
- Give Antigravity the complete CrimeNexus specification.
- Generate PostgreSQL and Neo4j schemas.
- Generate interconnected synthetic cases and records.
- Create 2–3 sample evidence files for upload/extraction.
- Test the upload → extraction → database → graph pipeline.

Person 2 — Website Prototype

- Build login and regional access UI.
- Build dashboard and Cases pages.
- Build upload flow.
- Connect populated data to case pages.
- Build graph and expandable cards.
- Implement timeline, Copilot and investigation workflow prototype.

You — Final Integration + Antigravity Specification

- Finalize the master technical specification.
- Standardize database, graph and API contracts.
- Integrate the prototype with the final architecture.
- Finalize AI Investigation Reconstruction.
- Finalize SHA-256 / Hyperledger workflow.
- Prepare deployment/demo configuration and final repository structure.

Other 3 Members — PPT / Presentation

- Person 3: Problem, existing gap, solution and differentiation.
- Person 4: Architecture, AI, graph, database, security, SHA-256 and blockchain.
- Person 5: Website screens, Investigation Reconstruction, Copilot, evidence verification, innovation and demo flow.

20. Final Demo Flow

```
1. REGION LOGIN
↓
2. DASHBOARD
↓
3. OPEN CASE #018
↓
4. UPLOAD FIR / CDR / TRANSACTION / CYBER DATA
↓
5. EXTRACT & RESOLVE ENTITIES
↓
6. BUILD KNOWLEDGE GRAPH
↓
7. START AI INVESTIGATION RECONSTRUCTION
↓
8. NARRATED EVENTS + LIVE GRAPH
↓
9. CLICK ENTITY → EXPANDABLE CARD
↓
10. DISCOVER CROSS-CASE CONNECTION
↓
11. SHOW NETWORK + TIMELINE + SIMILAR CASES
↓
12. FLAG EVIDENCE GAPS / CONTRADICTIONS
↓
13. ASK AI COPILOT QUESTIONS
↓
14. OPEN SUPPORTING EVIDENCE
↓
15. SHOW POTENTIALLY RELEVANT LEGAL PROVISIONS
↓
16. GENERATE DIGITAL EVIDENCE DOSSIER
↓
17. VERIFY SHA-256
↓
18. SHOW BLOCKCHAIN / AUDIT STATUS
```

21. Core Principle

CrimeNexus does not tell investigators who is guilty. It shows how evidence is connected, why a relationship matters, what happened over time, what remains unresolved, and where investigators may need to look next.

The system should distinguish between established evidence, analytical findings, unresolved questions, suggested review actions and potentially relevant legal provisions.